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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2007 question paper

5014 ENVIRONMENTAL MANAGEMENT

5014/02 Paper 2, maximum raw mark 60

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TANZANIA

1 (a) (i) S1 extinction inevitable; as no spray; S2 species saved; but not in natural habitat; S3 some may survive in habitat; [max

(ii) a proper environmental survey/eq; [1]

(iii) yes, qualified; or no, qualified; R refs to tourism [1]

(b) (i) not enough left to breed; so none can be exported in future/eq; [2]

(ii) B; less deaths (equal deaths) 0; more juveniles; increased breeding rate; AVP e.g. no need to supply special diet [3]

(c) (i) axes labelled; orientation; correct plots;; [4]

(ii) **X** faster growth Jan-June; difference bigger towards end/difference is 35 mm/rate of growth decreased in **Y** during May-June; [2]

(iii) natural vegetation and cabbage; [1]

(iv) to prevent disease/described; AVP: a cleaner environment/eq [1]

(v) prevent inbreeding/spread of wrong genes; increase vigour to offspring; maintain gene pool; adds biological variation; [2]

A ref to (increased/maintain) fertility

2 (a) ADV high yield in small area; everyone involved; organic fertiliser used; less chance of erosion; DIS labour intensive; only some of ground cultivated; [max 2]

(b) (i) pH; rainfall; aspect; slope angle; species of plant; soil type; AVP [max 2]

(ii) count all cobs and weigh some, calculate total weight; [1]

(iii) divide part **(ii)**/yield by number of square metres; [1]

(c) (i) some plants died of disease; lack of water; lack of nutrients; AVP; [2]

(ii) 750/3000 x 100 = 25%;; [2]

(d) (i) Surface runoff; so soil erosion; other ridges broken; top soil to bottom of field; AVP; e.g. crops damaged/destroyed [2]

(ii) lower yield due to low nutrients/eq; loss of soil structure; AVP;; e.g. less money for seeds [2]

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3 **(a)** **(i)** poison/problems in lungs; via skin; in organs; death;

food chain poisoned; so wild food not safe to eat; AVP;;

(ii) some fields now in poor condition; less food than before as less farming; less money to buy food/drugs; loss of farming skills; miners leave to mine other areas; AVP;; e.g. migration of other people [3]

(b) Use for HEP/electricity; make irrigated fields; source of clean water; detail of any use; [2]

(c) **(i)** same age; type of goat; size; type of enclosure; number of goats [2]

(ii) weigh goats each month; measure milk yield; health checks; AVP;; [3]

(iii) table drawn; headings; units; [3]

(d) **(i)** correct drawing; [1]

(ii) 15; [1]

(e) 1 x 365/2 = 182.5 ; 3000/182.5 = 16 goats; Allow 3000/365 = 8; A round up to 9 (allow working using 182 or 183) [2]

(f) **(i)** small fields; so high level of care; fits in with child care/other work; own income; [2]

(ii) Three valid arguments;;; each idea can have one development mark e.g. could use insecticide on own crops and so save the villagers' money (2mks) [3]

[TOTAL: 60]